

ELECTROMAGNETIC ENGINEERING —
COURSE PROJECT

ON

Resonant Wireless Power Transfer at 6.78 MHz MHz ISM Band

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6.78 MHz

ISM band operating frequency

$k = 0.075$

Target bifurcation threshold

$> 90\%$

Efficiency at operating point

50Ω

Achievable impedance match

Problem Statement & Motivation

Why wireless power transfer?

1

Consumer Electronics

Smartphones, wearables — galvanic isolation, waterproofing, convenience at mm-scale gaps (3–5 mm).

2

Medical Implants

Transcutaneous energy transfer to pacemakers, cochlear implants — no skin-penetrating connectors.

3

EV Charging

Under-vehicle pads at cm-scale gaps — enables autonomous parking-and-charge without human intervention.

Why 6.78 MHz ISM Band?

Globally unlicensed — no regulatory fragmentation. High enough frequency for compact resonant coils (small L), low enough to avoid significant tissue absorption. Standard adopted by the **AirFuel Alliance** for resonant WPT.

The Core Engineering Challenge

At MHz frequencies, tight inductive coupling ($k \rightarrow 1$) causes massive reflected impedance that collapses primary current — **maximum efficiency yields zero power**. The optimal operating regime must be found analytically and validated by simulation.

Project goal: Identify, simulate and validate the optimal coupling regime for a 6.78 MHz resonant WPT system targeting consumer electronics.

System Overview & Parameters

Series-Series Resonant Topology

Two magnetically coupled LC tanks, both resonated at ω_0 . Compensation capacitors cancel reactive impedance at resonance, maximising real power transfer.

$$\omega_0 = 1/\sqrt{LC}$$

$$M = k\sqrt{L_p \cdot L_s}$$

Mesh Equation Model

$$\begin{bmatrix} Z_p & -Z_m \\ -Z_m & Z_s \end{bmatrix} \begin{bmatrix} I_p \\ I_s \end{bmatrix} = \begin{bmatrix} V_{in} \\ 0 \end{bmatrix}$$
$$Z_p = R_p + j\omega L_p + 1/j\omega C_p$$
$$Z_s = (R_s + R_L) + j\omega L_s + 1/j\omega C_s$$

Simulation Parameters

Parameter	Value
Resonant Frequency f_0	6.78 MHz
Primary Inductance L_p	10 μ H
Secondary Inductance L_s	10 μ H
Coil ESR $R_p = R_s$	0.5 Ω
Load Resistance R_L	50 Ω
Input Voltage V_{in}	10 V
Target Coupling k	0.075

Compensation caps: $C_p = C_s = 1/(\omega_0^2 L)$ cancel reactive impedance at exactly 6.78 MHz.

Simulation Methodology

WPT.m — Resonant Performance

Mesh equation solver over 5–8 MHz frequency sweep (1000 points). Computes efficiency, voltage gain and input impedance phase for $k = 0.075$.

Output: Resonant peaks, phase zero-crossing, crossing, gain rolloff

WPT_power_plot.m — Coupling Regimes

Four coupling values ($k = 0.011, 0.075, 0.075, 0.117, 0.20$) swept 6–7.5 MHz. Gain, load power and efficiency computed. Reflection coefficient Γ mapped onto Smith Chart.

Output: Bifurcation, flat-top plateau, $50\ \Omega$ trajectory

WPT_error_plot.m & WPT_relations.m

Neumann integral (100×100 pt mesh) for lateral offset (0–15 cm) and angular tilt (0–90°). Geometric approximation for axial distance and turn-count scaling.

Output: Flux null point, $1/z^3$ decay, invariance of k to N

WPT_rectenna.m — Transient Time-Domain

Microsecond-by-microsecond differential solver over a $30\ \mu\text{s}$ window. Models a non-linear Full-Bridge Schottky rectifier and an LC choke filter ($1\ \mu\text{H} / 2.2\ \mu\text{F}$). Computes true active power via instantaneous voltage-current integration.

Output: LC overshoot, Continuous Conduction Mode (CCM) stabilization, DC ripple voltage, and true rectification efficiency.

MATLAB

Simulation environment

1,000 pts

Per frequency sweep

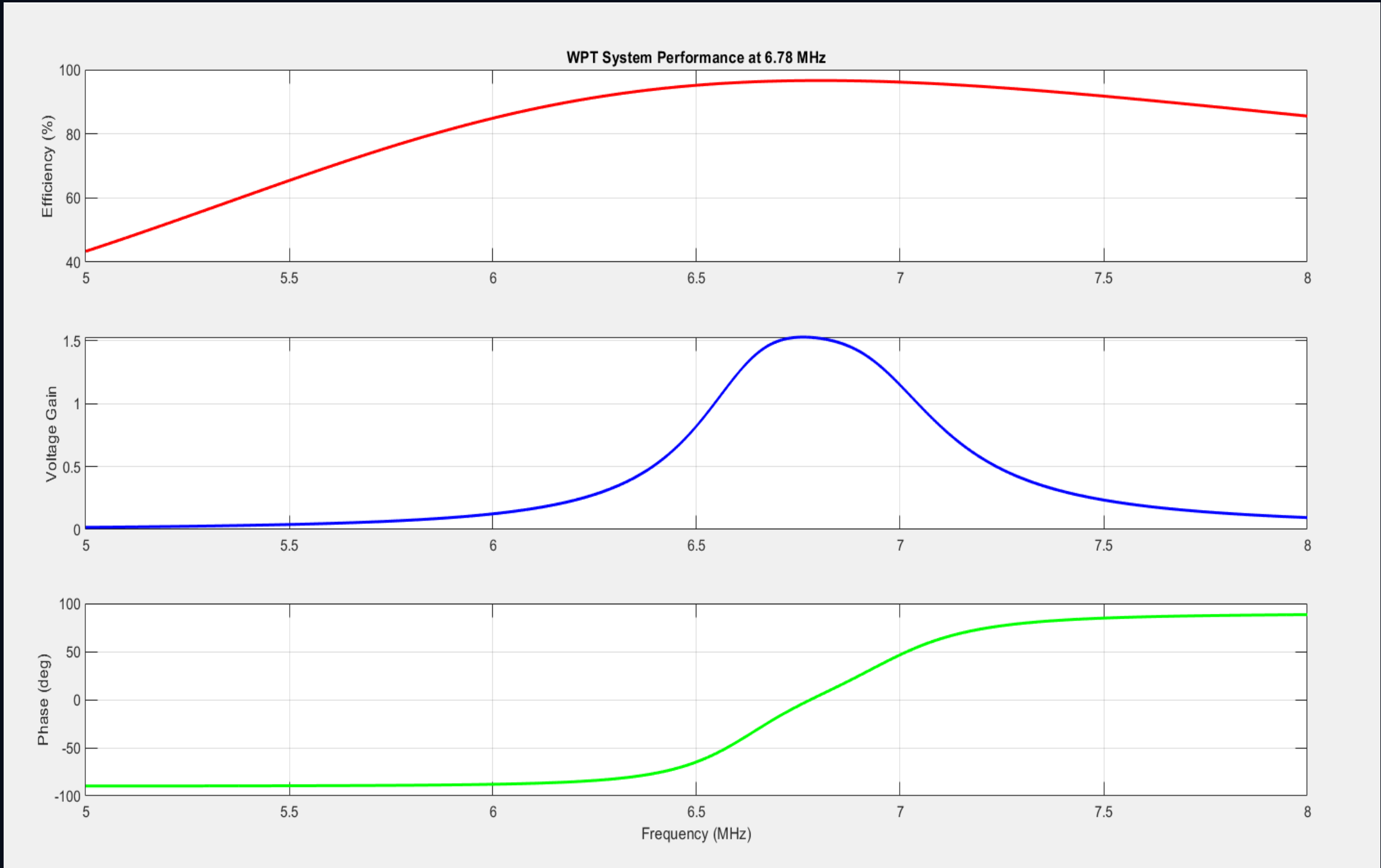
100×100

Neumann mesh resolution

5 scripts

Independent analysis modules

System Performance at Resonance



96.4% Peak efficiency at exactly 6.78 MHz

— efficiency is sharply tuned to the resonant frequency.

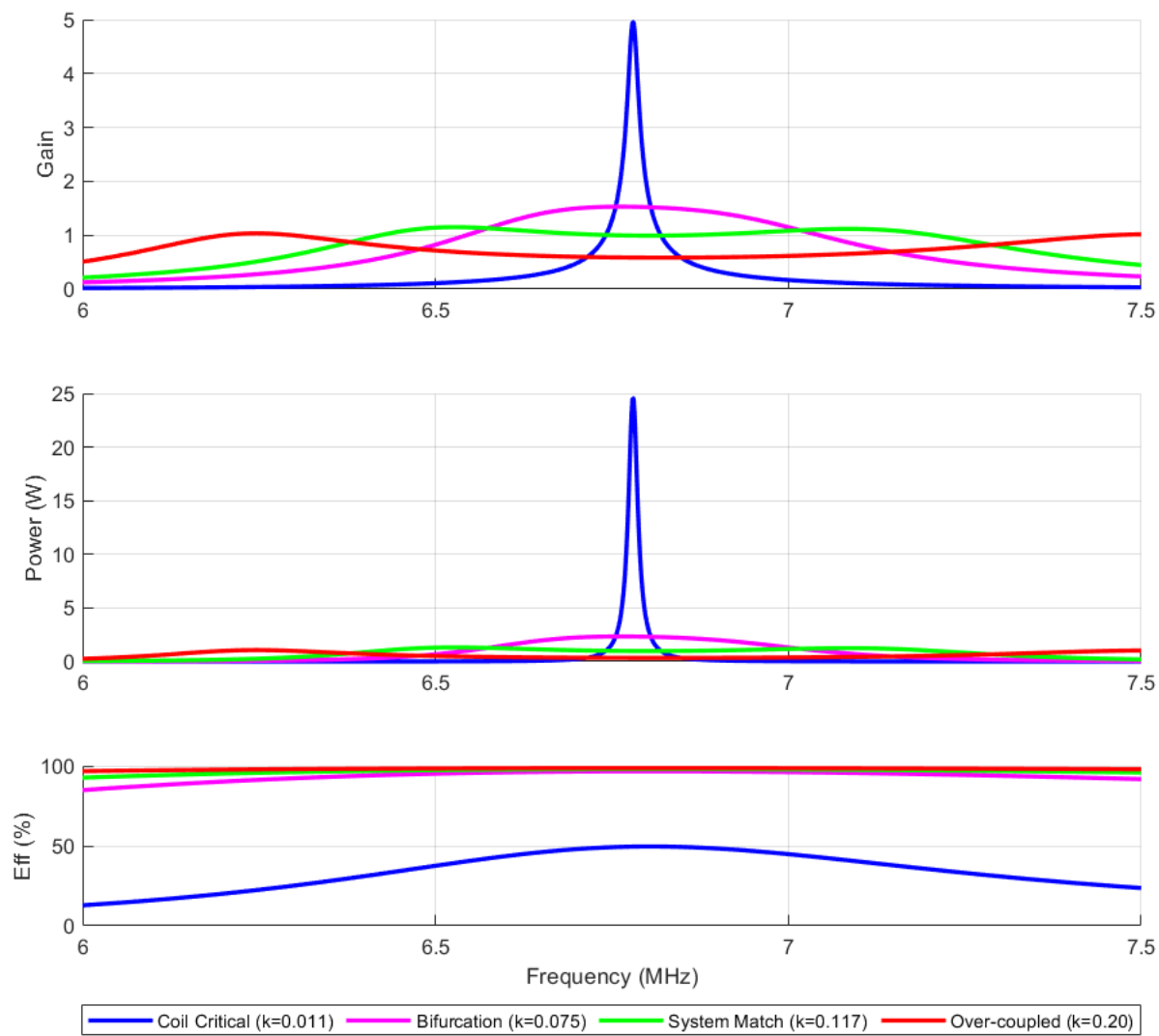
Voltage gain peaks at resonance then rolls off symmetrically — confirms correct series-series compensation.

Phase crosses zero at exactly f_0 — the system presents a purely resistive input impedance at the design frequency.

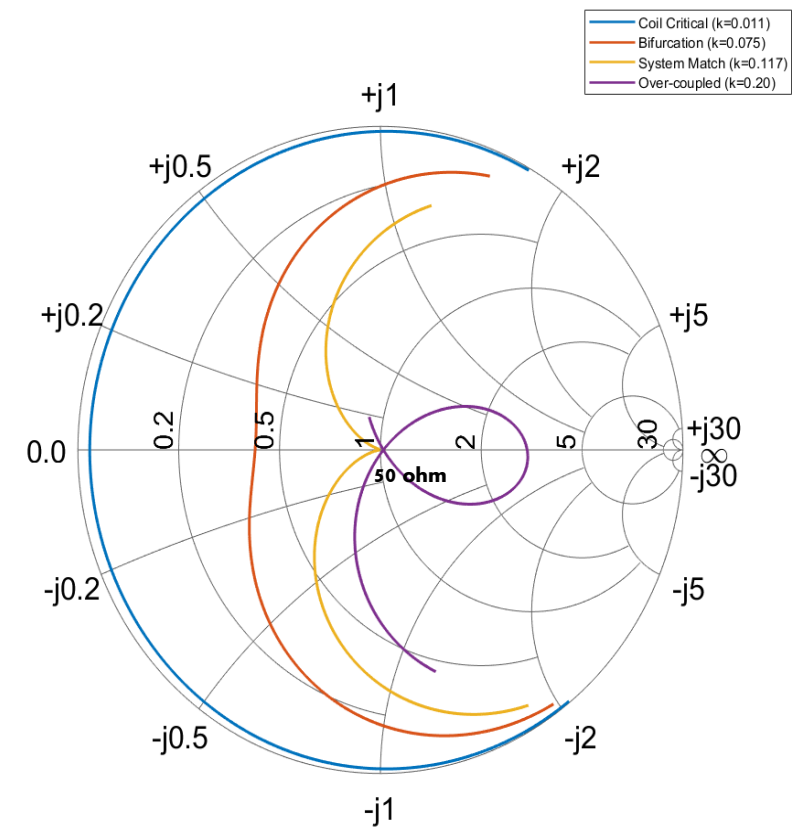
Key confirmation: Simulation validates that series-series compensation correctly resonates both tanks at the target ISM frequency.

Coupling Regimes & Bifurcation

Gain · Power · Efficiency vs. Frequency



Impedance Trajectory (Smith Chart)



— $k = 0.011$ — sharp fragile peak

— $k = 0.075$ — flat-top plateau ★

— $k = 0.117$ — perfect $50\ \Omega$ match

— $k = 0.20$ — over-coupled (split)

$k = 0.075$ produces a **flat-top power plateau** — spatial freedom for the user.

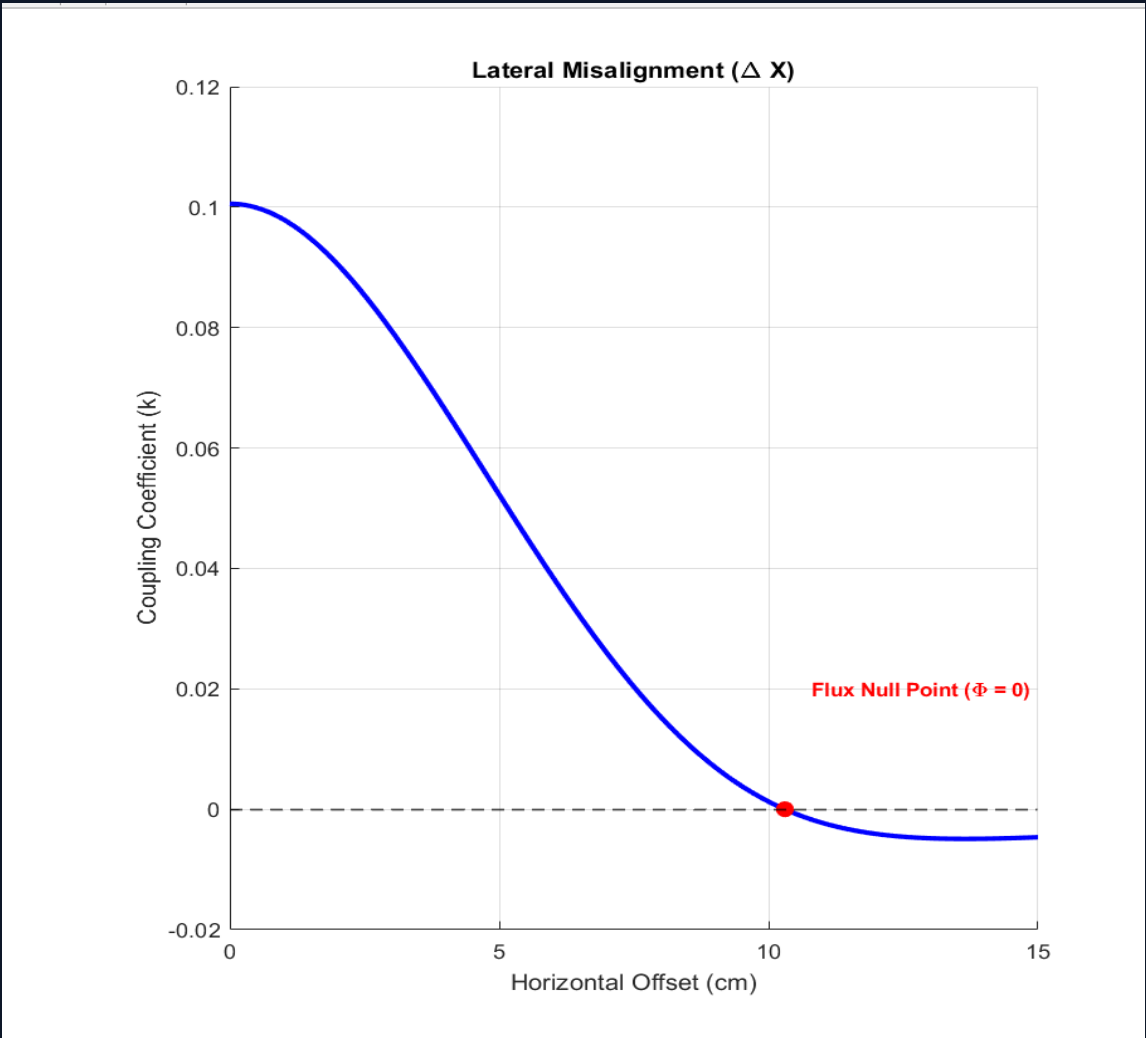
$k = 0.117$ Smith Chart trajectory passes through $\Gamma = 0$ — perfect $50\ \Omega$ match via geometry alone.

$k = 0.20$ — large Smith loop = severe reactive detuning. Inverter thermal risk.

IMN required to rotate $k = 0.075$ to $50\ \Omega$ centre. Lower peak power solved by scaling **Vin**.

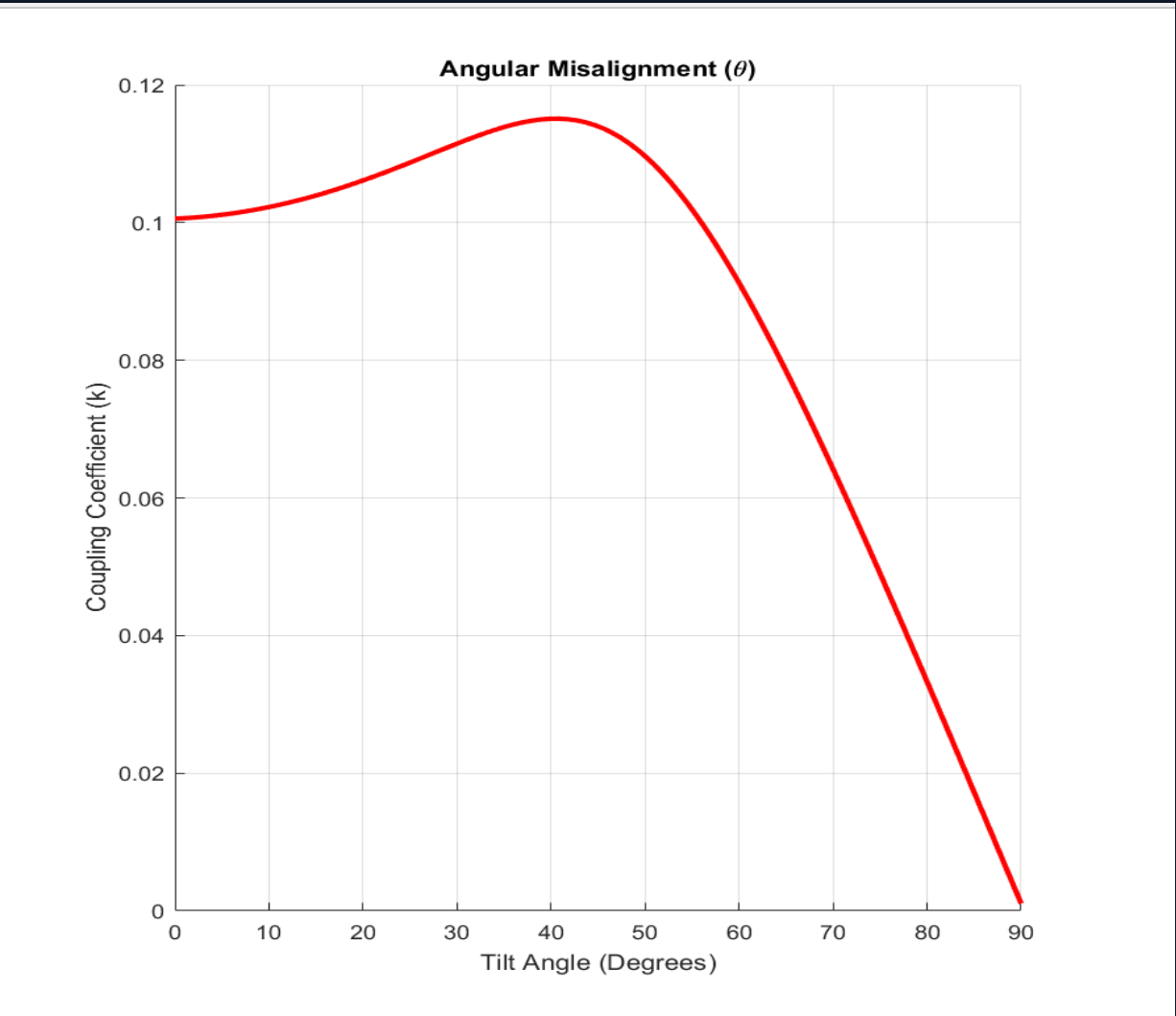
Spatial Misalignment Analysis

Lateral offset ΔX — Neumann integral ($R = 5\text{ cm}$, $z\text{-gap} = 5\text{ cm}$)



Flux Null at $\approx 11\text{ cm}$ lateral offset: Field lines entering and exiting the same aperture cancel — net $\Phi = 0$. k drops to zero. Validated by Neumann integration.

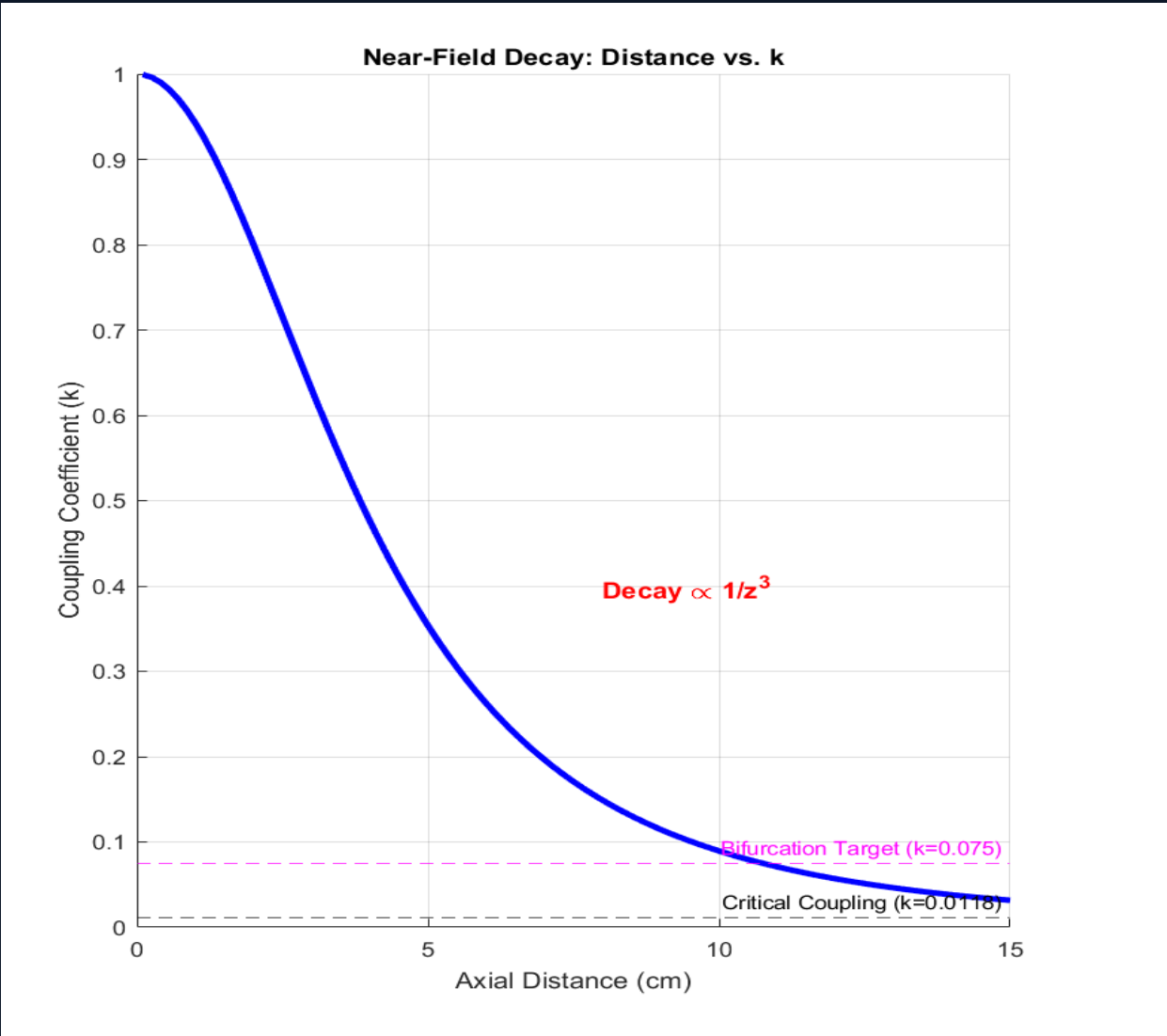
Angular tilt θ — Rx coil rotated about the Y-axis



k degrades **monotonically** with tilt, falling below the $k = 0.075$ target at $\approx 45^\circ$. Practical tolerance: tilt $< 30^\circ$ for stable operation.

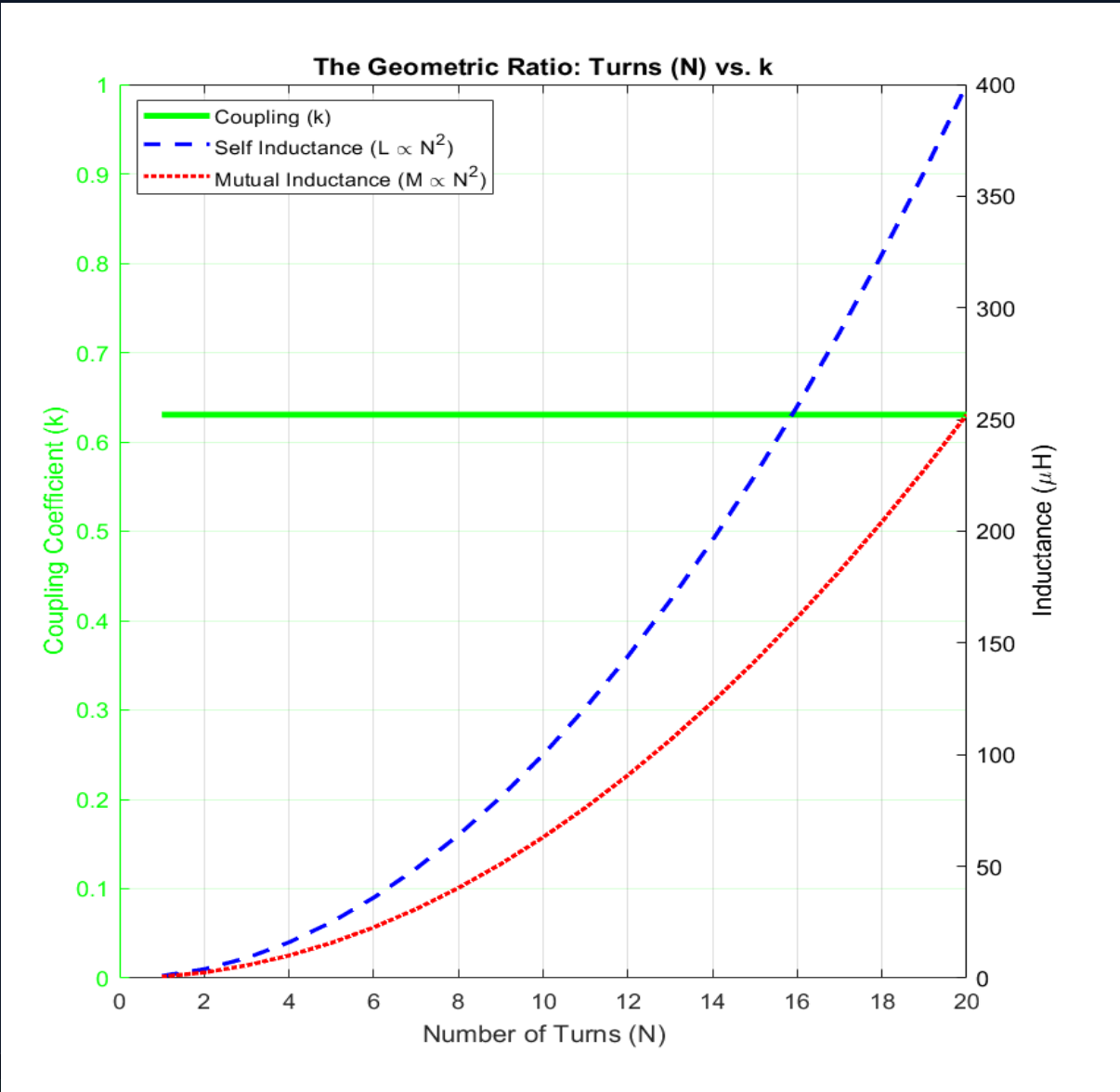
Distance & Turn-Count Relationships

Axial distance z vs. coupling coefficient ($R = 5\text{ cm}$ coils)



Coupling decays $\propto 1/z^3$ — characteristic near-field behaviour. At 3 mm only asymmetric topology achieves $k = 0.075$ without over-coupling.

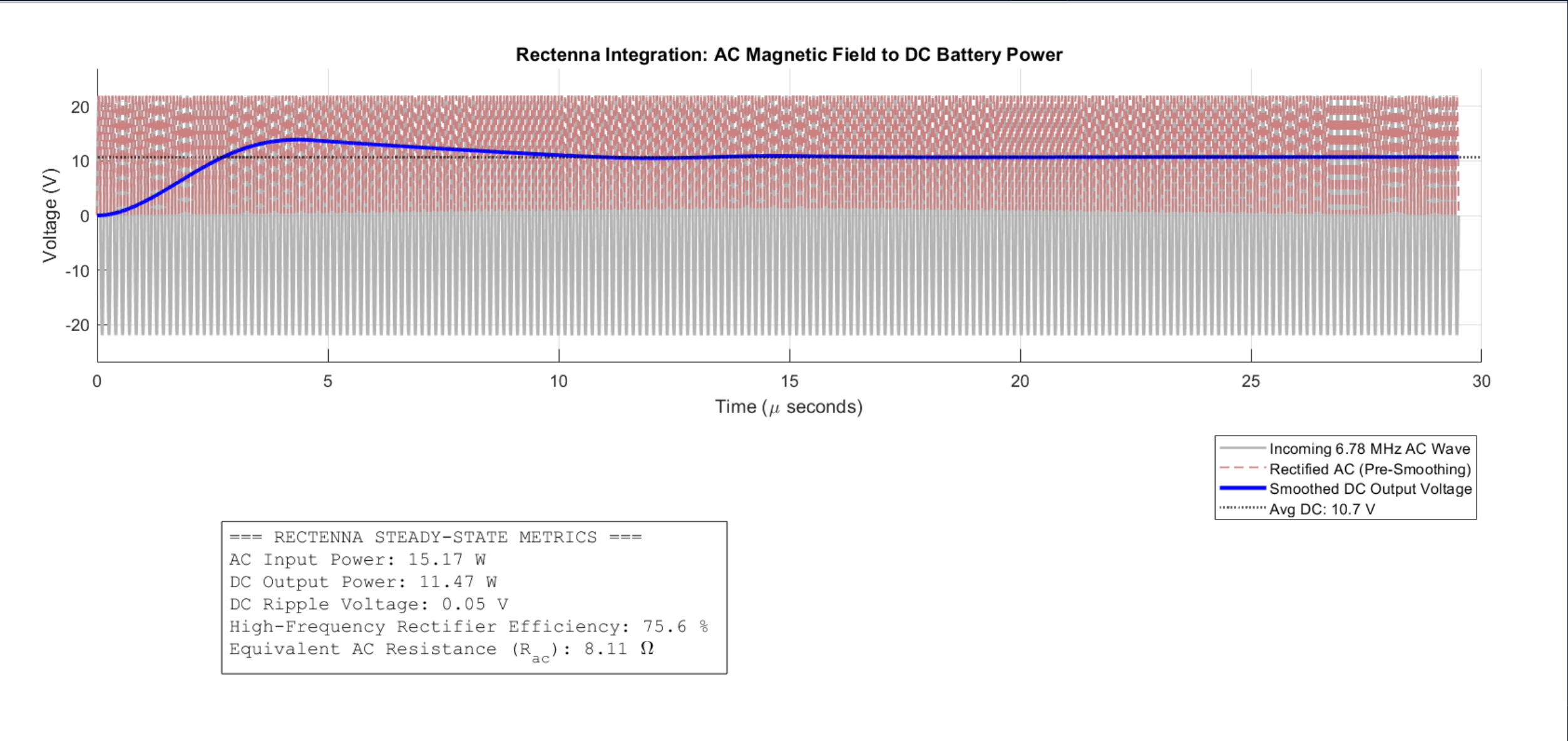
Turn count N vs. k & inductance ($z = 3\text{ cm}$ fixed)



Design insight: Adding turns does *not* improve k — both M and L scale as N^2 , so $k = M/\sqrt{LL}$ remains constant. More turns only increases inductance.

Rectenna Integration: AC to DC rectification

The Objective: Convert the 6.78 MHz AC magnetic wave into stable DC power for a 14.8V Battery Management System (BMS)



$$\text{Ripple \%} = \left(\frac{V_{max} - V_{min}}{V_{DC}} \right) \times 100$$

High-Frequency Full-Bridge Schottky Rectifier
with an LC Filter.

The Outcome: Continuous Conduction Mode (CCM) achieved in ~15
micro-seconds, delivering 11.4 Watts of stable DC power at 75.6%
efficiency with a 0.05 V ripple.

Key Takeaways

①

Optimal operating point: $k = 0.075$. The gain bifurcation threshold delivers a flat-top power plateau with $> 90\%$ efficiency — balancing power and spatial freedom.

②

$50\ \Omega$ match achievable at $k = 0.117$ purely through coil positioning — verified on Smith Chart. An IMN rotates the $k = 0.075$ point to centre.

③

Lower plateau power solved by V_{in} scaling — the coil geometry is unchanged. This is the correct engineering lever.

④

Misalignment tolerance: lateral $< \sim 8$ cm (before flux null at ≈ 11 cm), angular $< 30^\circ$ for stable operation above $k = 0.075$.

⑤

Adding turns does not improve k . Both M and L scale as N^2 — only inductance changes. Geometry (radius, gap) is the only lever for k .

⑥

Coupling decays $\propto 1/z^3$ — near-field regime. Fill factor $\rho = 0.4$ maximises Q by eliminating inner turns that add resistance without contributing mutual flux.

REFERENCES

- ① [1] Z. Wang, "Efficiency Analysis of S-S Resonant Wireless Power Transfer Systems: Impacts of Coupling Coefficient, Q-Factor, and ESR," in Proc. 2nd Int. Conf. Mech., Elec. Eng. & Auto. (ICMEEA 2025), 2025, doi:10.2991/978-94-6463-821-9_111.
- ② [2] M. M. Taib and A. Ponniran, "Optimization of CC-Coil Design for Wireless Power Transfer System with Series-Series Magnetic Resonance Compensation Technique," J. Electron. Volt. Appl., vol. 2, no. 2, 2021.
- ③ [3] L. Gu, G. Zulauf, A. Stein, P. A. Kyaw, T. Chen, and J. M. Rivas-Davila, "Design and Optimization of 6.78 MHz Wireless Power Transfer with Self-Resonant Coils," COMPEL, 2020, doi:10.1109/COMPEL49091.2020.9265744.

Conclusion

What the simulation validated

Mesh-equation MATLAB model **correctly predicts all impedance trajectories**, resonant peaks, and power limits for a 6.78 MHz series-series WPT system.

The **Neumann integral confirms the flux null point** at ≈ 11 cm lateral offset and quantifies the angular tolerance limit.

Simulation provides a **validated design baseline** for physical hardware — all key operating parameters (k , f_o , Z in Ω) confirmed analytically.

The 6.78 MHz ISM band, $k = 0.075$ bifurcation threshold, and series-series compensation constitute a robust, validated design framework for consumer WPT.